

Q 18, 19 - Solving trig equations

Theory

Using trig identities, it is easy to solve trig equations like these:

$$(1) \quad 4 \cos^2 \theta - 4 \sin \theta = 1$$

$$(2) \quad \cos(2\theta) + \sin \theta = 0$$

🔥 Convert to polynomial

Notice in (1) the mismatch of types: $\cos \theta$ in the square but $\sin \theta$ without the square.

Notice in (2) the mismatch of inputs: 2θ in the cosine and θ in the sine.

Method: In either situation, convert to a single *type* and a single *input* using trig identities.

The result has the form of polynomial equation that can be solved.

≡ Examples

Example (1):

$$4 \cos^2 \theta - 4 \sin \theta = 1 \quad \ggg \quad 4(1 - \sin^2 \theta) - 4 \sin \theta = 1$$

$$\overset{X=\sin \theta}{\ggg} \quad 4(1 - X^2) - 4X = 1$$

$$\ggg \quad -4X^2 - 4X + 3 = 0$$

$$\ggg \quad X = -\frac{3}{2}, \frac{1}{2}$$

$$\ggg \quad \theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

Example (2):

$$\cos(2\theta) + \sin \theta = 0 \quad \ggg \quad \cos^2 \theta - \sin^2 \theta + \sin \theta = 0$$

$$\ggg \quad 1 - 2 \sin^2 \theta + \sin \theta = 0$$

$$\ggg \quad -2X^2 + X + 1 = 0$$

$$\ggg \quad X = -\frac{1}{2}, 1$$

$$\ggg \quad \theta = \frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

Problems

- Solve these equations and verify your answers.
- If your answer is wrong, try to find your mistake and fix your solution.
- Afterwards, study the full solution provided.

Assume that x and θ lie in $[0, 2\pi)$ for all these problems.

1. $\cos(2x) - 2 \cos x + 1 = 0$

Answer: $x = 0, \frac{\pi}{2}, \frac{3\pi}{2}$

Solution >

Use $\cos(2x) = 2 \cos^2 x - 1$:

$$\cos(2x) - 2 \cos x + 1 = 0 \quad \ggg \quad 2 \cos^2 x - 2 \cos x = 0$$

Substitute $X = \cos x$ and solve:

$$\ggg \quad 2X^2 - 2X = 0 \quad \ggg \quad 2X(X - 1) = 0 \quad \ggg \quad X = 0, 1$$

$$\bullet \quad \cos x = 1 \quad \ggg \quad x = 0$$

$$\bullet \quad \cos x = 0 \quad \ggg \quad x = \frac{\pi}{2}, \frac{3\pi}{2}$$

2. $\sin(2x) + \sin x = 0$

Answer: $x = 0, \pi, \frac{2\pi}{3}, \frac{4\pi}{3}$

 Solution >

Use $\sin(2x) = 2 \sin x \cos x$:

$$2 \sin x \cos x + \sin x = \sin x (2 \cos x + 1) = 0$$

So $\sin x = 0$ or $2 \cos x + 1 = 0$.

- $\sin x = 0 \quad \gg \gg \quad x = 0, \pi$
- $\cos x = -\frac{1}{2} \quad \gg \gg \quad x = \frac{2\pi}{3}, \frac{4\pi}{3}$

3. $\cos(2x) + \cos x + 1 = 0$

Answer: $x = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{2\pi}{3}, \frac{4\pi}{3}$

 Solution >

Use $\cos(2x) = 2 \cos^2 x - 1$. Substitute $X = \cos x$ and solve:

$$\begin{aligned} \cos(2x) + \cos x + 1 = 0 & \gg \gg 2 \cos^2 x - 1 + \cos x + 1 \\ & \gg \gg 2X^2 + X = 0 \\ & \gg \gg X(2X + 1) = 0 \end{aligned}$$

So $X = 0$ or $X = -\frac{1}{2}$.

- $X = 0 \quad \gg \gg \quad x = \frac{\pi}{2}, \frac{3\pi}{2}$
- $X = -\frac{1}{2} \quad \gg \gg \quad x = \frac{2\pi}{3}, \frac{4\pi}{3}$

4. $\cos(2x) + \frac{1}{2} = 0$

Answer: $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$

 Solution >

Use $\cos(2x) = 2\cos^2 x - 1$ and solve for $\cos x$:

$$\cos(2x) + \frac{1}{2} = 0 \quad \gg \gg \quad 2\cos^2 x - 1 + \frac{1}{2} = 0$$

$$\gg \gg \quad 2\cos^2 x - \frac{1}{2} = 0$$

$$\gg \gg \quad \cos^2 x = \frac{1}{4}$$

$$\gg \gg \quad \cos x = \pm \frac{1}{2}$$

- $\cos x = \frac{1}{2} \quad \gg \gg \quad x = \frac{\pi}{3}, \frac{5\pi}{3}$
- $\cos x = -\frac{1}{2} \quad \gg \gg \quad x = \frac{2\pi}{3}, \frac{4\pi}{3}$

5.

$$\tan(2\theta) - 3\tan\theta = 0$$

Answer: $\theta = 0, \frac{\pi}{6}, \frac{5\pi}{6}, \pi, \frac{7\pi}{6}, \frac{11\pi}{6}$

 Solution >

Use $\tan(2\theta) = \frac{2\tan\theta}{1-\tan^2\theta}$ and substitute $X = \tan\theta$:

$$\tan(2\theta) - 3\tan\theta = 0 \quad \gg \gg \quad \frac{2X}{1-X^2} - 3X = 0$$

$$\gg \gg \quad X\left(\frac{2}{1-X^2} - 3\right) = 0$$

Suppose that $X = 0$. Then $\theta = 0, \pi$.

Suppose that $\frac{2}{1-X^2} - 3 = 0$. Then:

$$\gg \gg \quad 2 = 3(1 - X^2) \quad \gg \gg \quad X^2 = \frac{1}{3} \quad \gg \gg \quad X = \pm \frac{1}{\sqrt{3}}$$

- $\tan \theta = \frac{1}{\sqrt{3}} \gg \gg \theta = \frac{\pi}{6}, \frac{7\pi}{6}$
- $\tan \theta = -\frac{1}{\sqrt{3}} \gg \gg \theta = \frac{5\pi}{6}, \frac{11\pi}{6}$

6. $\sin^2 \theta + \sin \theta - 2 = 0$

Answer: $\theta = \frac{\pi}{2}$

 **Solution** >

Substitute $X = \sin \theta$ and solve:

$$\gg \gg X^2 + X - 2 = 0 \gg \gg (X + 2)(X - 1) = 0$$

- $\sin \theta = -2 \gg \gg$ impossible
- $\sin \theta = 1 \gg \gg \theta = \frac{\pi}{2}$

7. $\sin(2\theta) = \cos \theta$

Answer: $\theta = \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{3\pi}{2}$

 **Solution** >

Use $\sin(2\theta) = 2 \sin \theta \cos \theta$:

$$2 \sin \theta \cos \theta = \cos \theta \gg \gg \cos \theta (2 \sin \theta - 1) = 0$$

- $\cos \theta = 0 \gg \gg \theta = \frac{\pi}{2}, \frac{3\pi}{2}$
- $\sin \theta = \frac{1}{2} \gg \gg \theta = \frac{\pi}{6}, \frac{5\pi}{6}$

8. $\cos(2\theta) - \sin \theta = 0$

Answer: $\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$

 Solution >

Use $\cos(2\theta) = 1 - 2\sin^2 \theta$ and substitute $X = \sin \theta$:

$$\ggg 1 - 2\sin^2 \theta - \sin \theta \ggg 2X^2 + X - 1 = 0$$

Solve this using the quadratic formula:

$$X = \frac{-1 \pm \sqrt{1+8}}{4} = \frac{-1 \pm 3}{4} = \frac{1}{2}, -1$$

- $\sin \theta = \frac{1}{2} \ggg \theta = \frac{\pi}{6}, \frac{5\pi}{6}$
- $\sin \theta = -1 \ggg \theta = \frac{3\pi}{2}$

9. $3\sec^2 \theta - 2\tan^2 \theta - 4 = 0$

Answer: $\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

 Solution >

Use $\sec^2 \theta = 1 + \tan^2 \theta$ and substitute $X = \tan \theta$:

$$3(1 + X^2) - 2X^2 - 4 = 0 \ggg X^2 - 1 = 0$$

Therefore $X = \pm 1$.

- $\tan \theta = 1 \ggg \theta = \frac{\pi}{4}, \frac{5\pi}{4}$
- $\tan \theta = -1 \ggg \theta = \frac{3\pi}{4}, \frac{7\pi}{4}$